

Functions and Function Notation (Part IV)

Domain: All of the inputs of a function (typically x-values) that are able to produce an output.

Range: All of the outputs of a function (typically y-values) that are produced from the inputs in the domain.

✱ Neither imaginary nor complex numbers are allowed to be inputs or outputs of functions.

Finding the Domain and Range of Functions Using Graphs

Example

Find the domain and range of the function below.

$$f(x) = \log_2 x$$

Domain

Reasoning: Both zero and negative numbers produce an undefined logarithmic expression.

Words: All real numbers greater than zero.

Set Notation: $\{x | x > 0\}$

Interval Notation: $(0, \infty)$

Logarithm Property III

$$\log_b b = 1$$

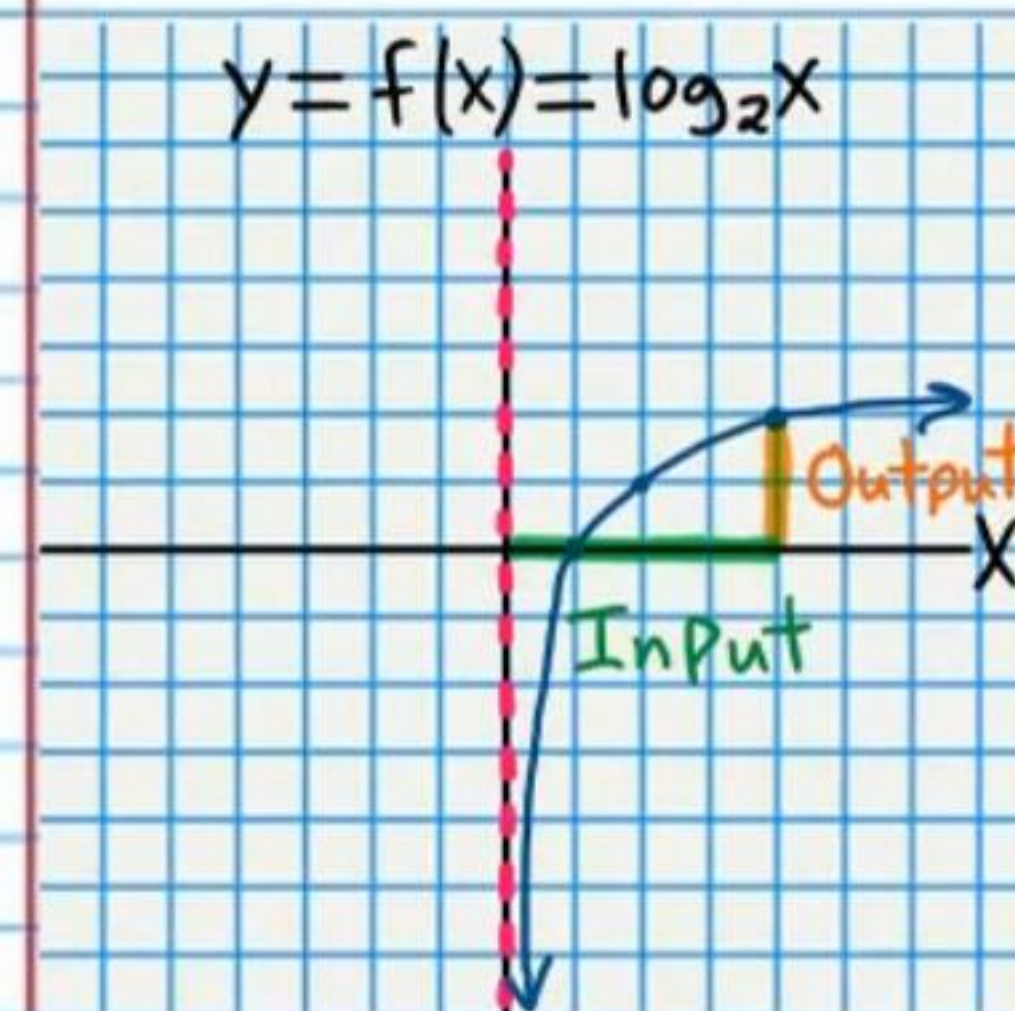
Logarithm Property II

$$\log_b b^x = x$$

Logarithm Property IX

$$\log_b 1 = 0$$

Range



Reasoning: The graph shows that its points can have any y-value (i.e. positive, negative, or zero), and the y-values increase or decrease indefinitely.

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

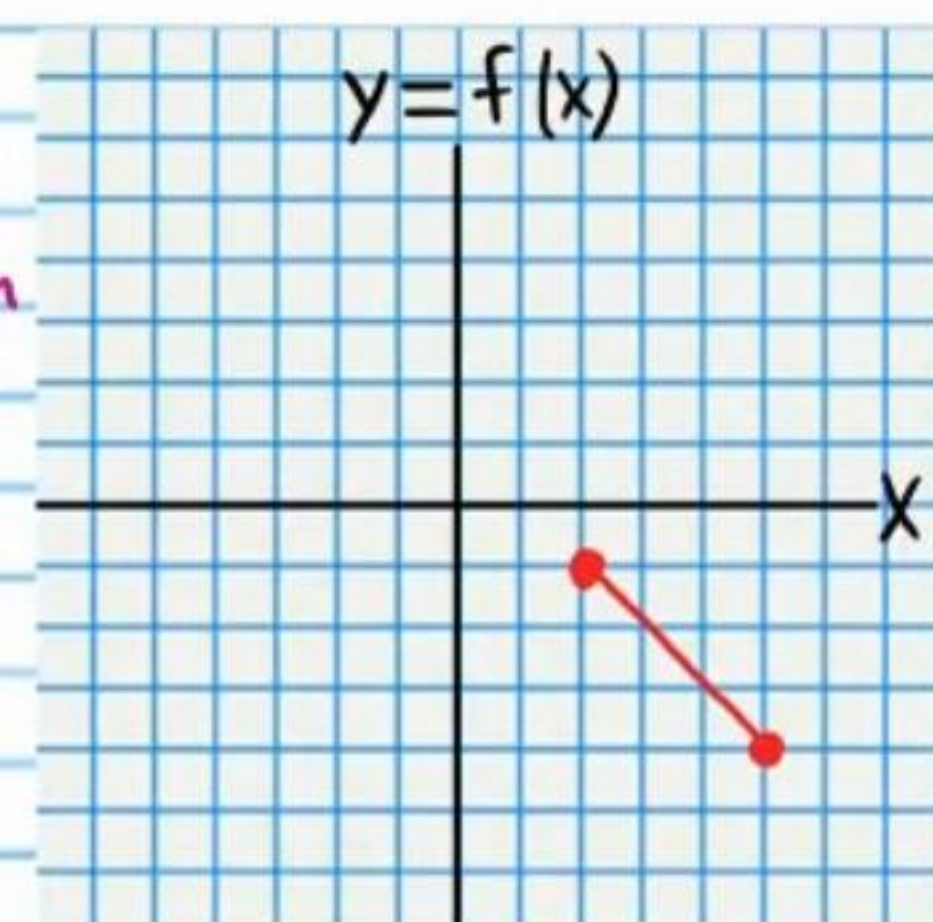
The graphs of various functions are written below. Write the domain and range of each function in words, set notation, and interval notation.

① Domain

Words: All real numbers between 2 and 5, inclusive.

Set Notation: $\{x | 2 \leq x \leq 5\}$

Interval Notation: $[2, 5]$



Range

Words: All real numbers between -4 and -1, inclusive.

Set Notation: $\{y | -4 \leq y \leq -1\}$

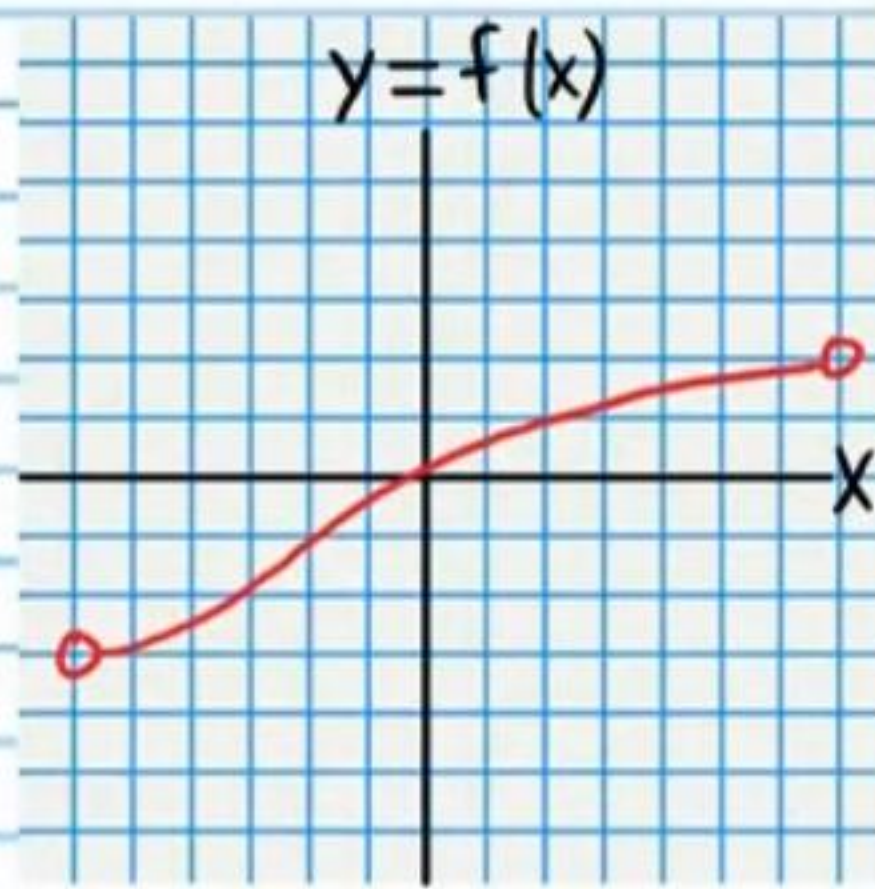
Interval Notation: $[-4, -1]$

② Domain

Words: All real numbers between -6 and 7.

Set Notation: $\{x \mid -6 < x < 7\}$

Interval Notation: $(-6, 7)$



Range

Words: All real numbers between -3 and 2.

Set Notation: $\{y \mid -3 < y < 2\}$

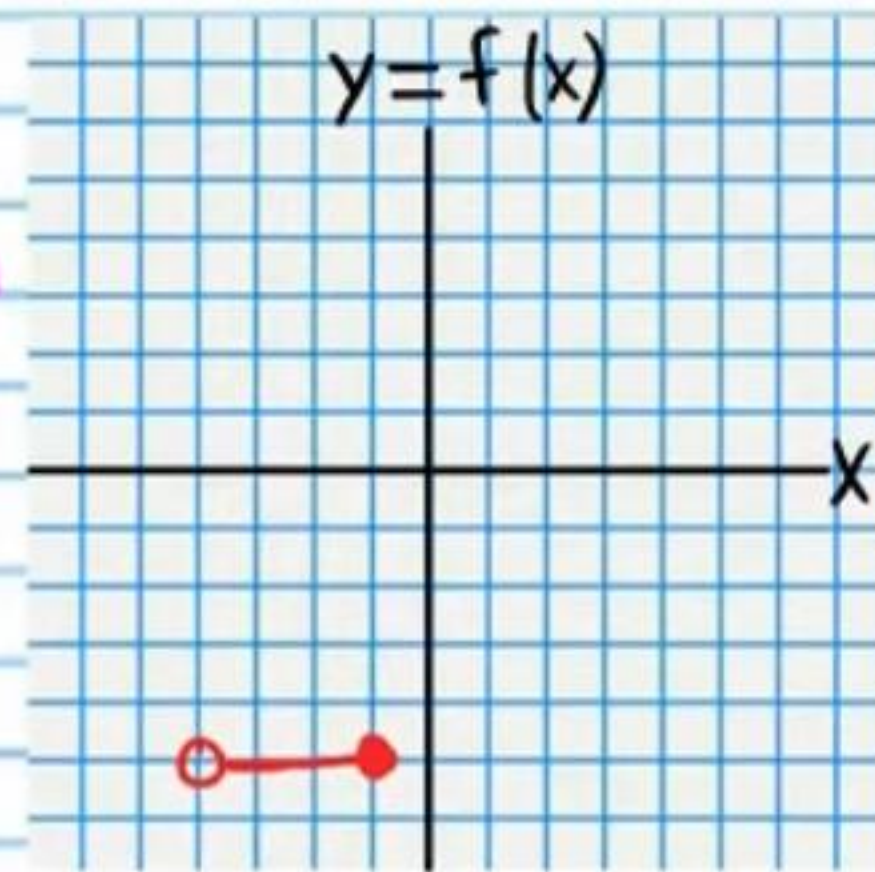
Interval Notation: $(-3, 2)$

③ Domain

Words: All real numbers between -4 and -1, including -1.

Set Notation: $\{x \mid -4 < x \leq -1\}$

Interval Notation: $(-4, -1]$



Range

Words: The only value in the range is -5.

Set Notation: $\{y \mid y = -5\}$

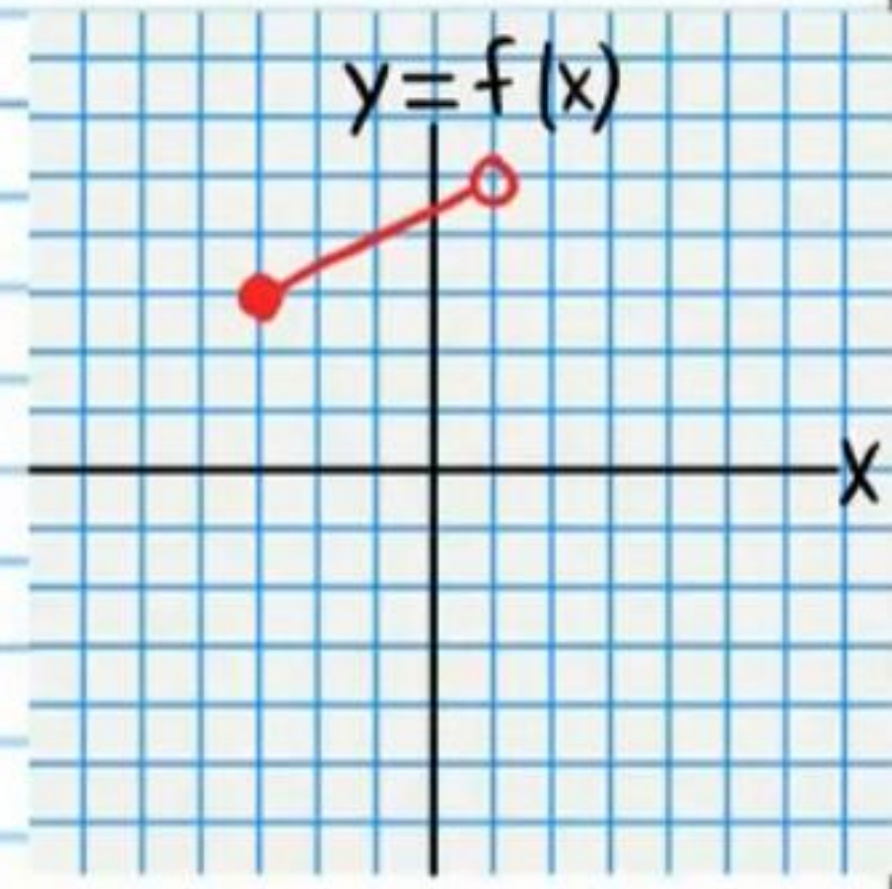
Interval Notation: $[-5, -5]$

④ Domain

Words: All real numbers between -3 and 1, including -3.

Set Notation: $\{x \mid -3 \leq x < 1\}$

Interval Notation: $[-3, 1)$



Range

Words: All real numbers between 3 and 5, including 3.

Set Notation: $\{y \mid 3 \leq y < 5\}$

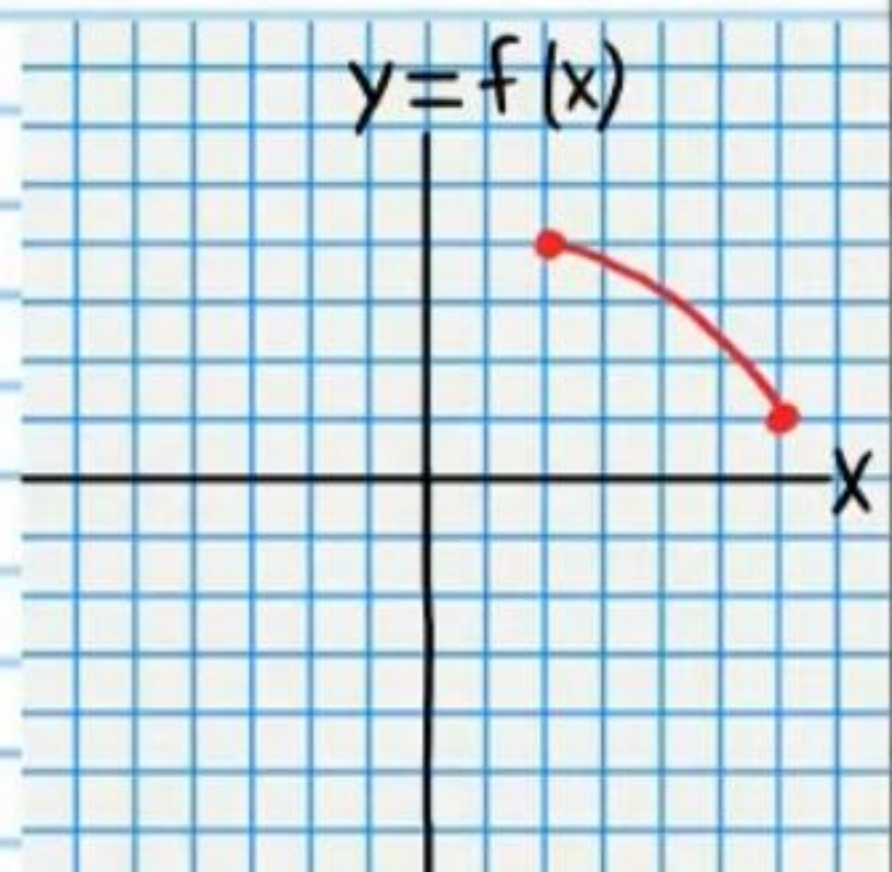
Interval Notation: $[3, 5)$

⑤ Domain

Words: All real numbers between 2 and 6, inclusive.

Set Notation: $\{x \mid 2 \leq x \leq 6\}$

Interval Notation: $[2, 6]$



Range

Words: All real numbers between 1 and 4, inclusive.

Set Notation: $\{y \mid 1 \leq y \leq 4\}$

Interval Notation: $[1, 4]$

⑥ Domain

Words: All real numbers between -5 and 2.

Set Notation: $\{x \mid -5 < x < 2\}$

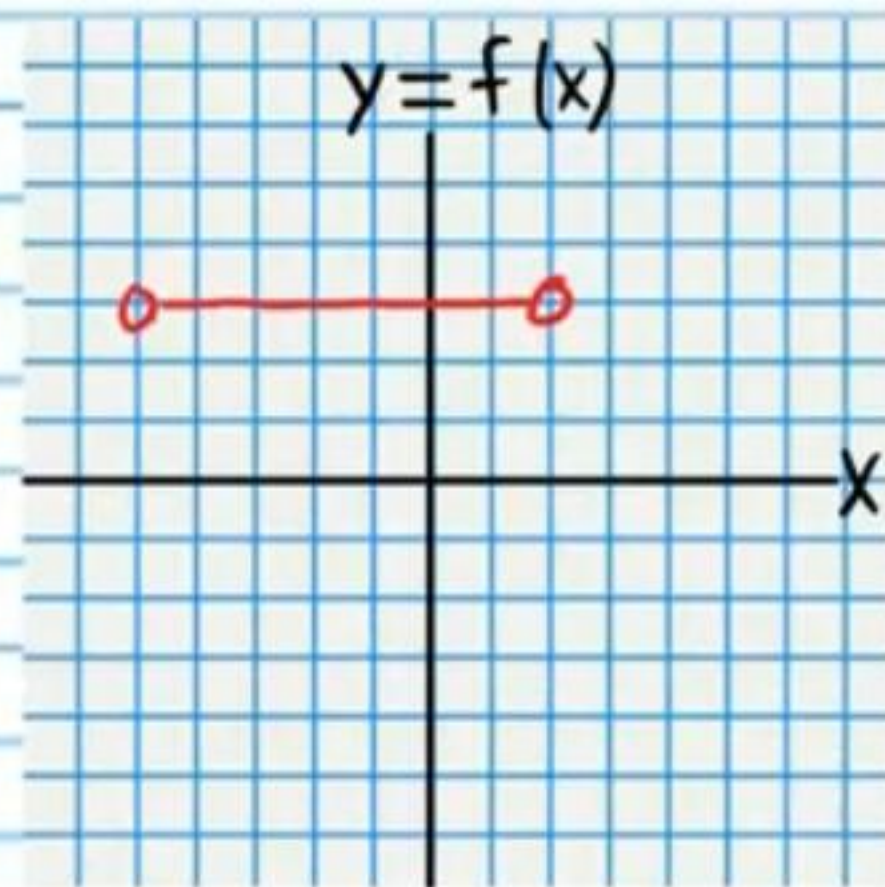
Interval Notation: $(-5, 2)$

Range

Words: The only value in the range is 3.

Set Notation: $\{y \mid y = 3\}$

Interval Notation: $[3, 3]$



⑦ Domain

Words: All real numbers between 1 and 6, including 1.

Set Notation: $\{x \mid 1 \leq x < 6\}$

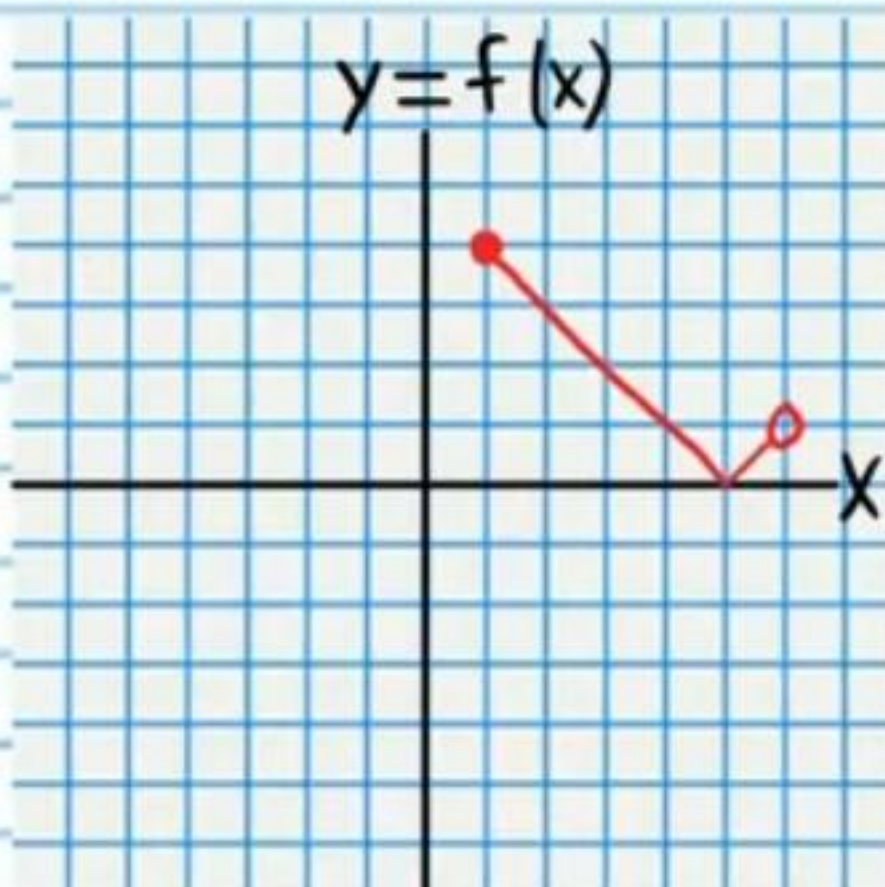
Interval Notation: $[1, 6)$

Range

Words: All real numbers between zero and 4, inclusive.

Set Notation: $\{y \mid 0 \leq y \leq 4\}$

Interval Notation: $[0, 4]$



⑧ Domain

Words: All real numbers between zero and 5, including 5.

Set Notation: $\{x \mid 0 \leq x \leq 5\}$

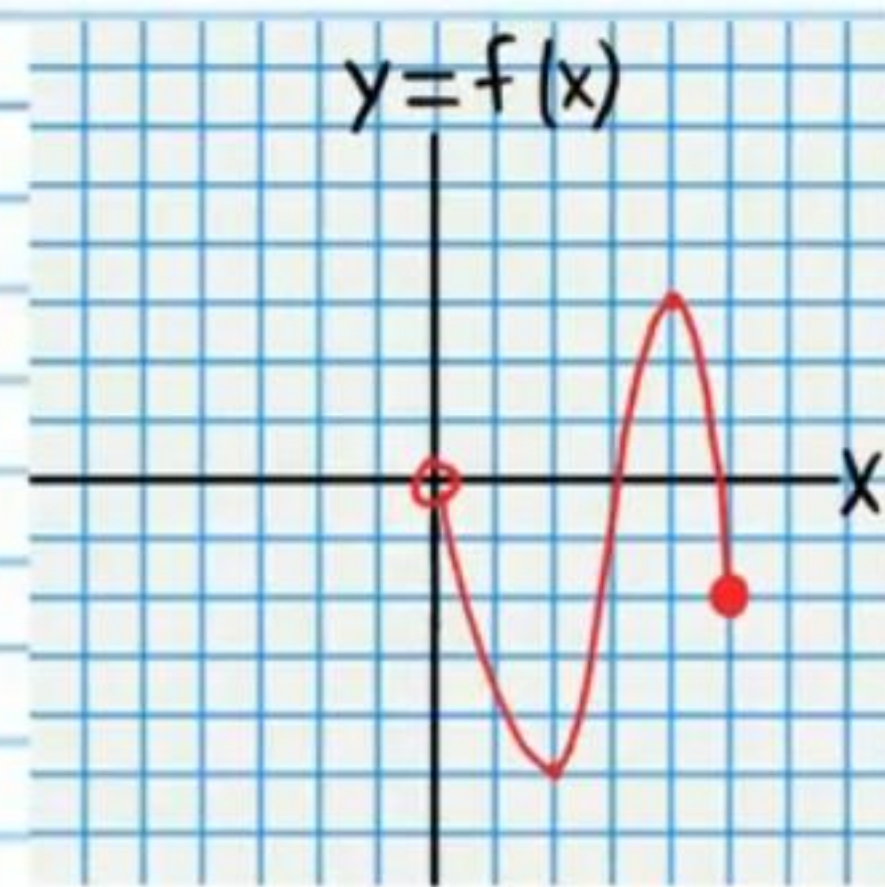
Interval Notation: $[0, 5]$

Range

Words: All real numbers between -5 and 3, inclusive.

Set Notation: $\{y \mid -5 \leq y \leq 3\}$

Interval Notation: $[-5, 3]$



⑨ Domain

Words: All real numbers between -4 and 4.

Set Notation: $\{x \mid -4 < x < 4\}$

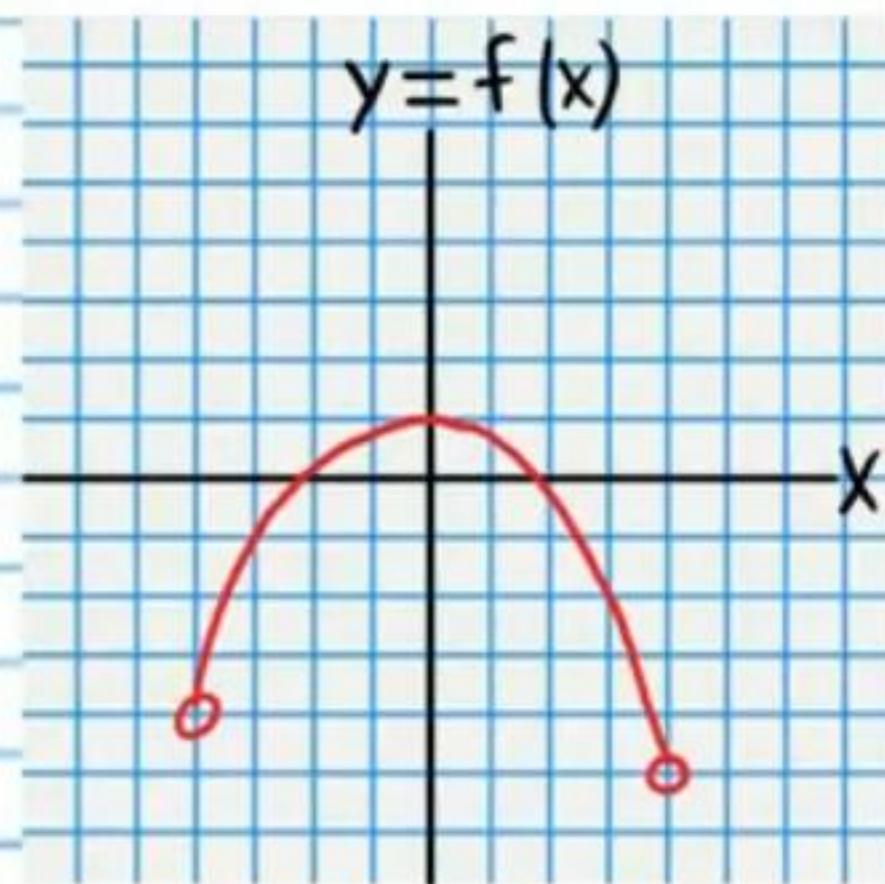
Interval Notation: $(-4, 4)$

Range

Words: All real numbers between -5 and 1, including 1.

Set Notation: $\{y \mid -5 \leq y \leq 1\}$

Interval Notation: $[-5, 1]$

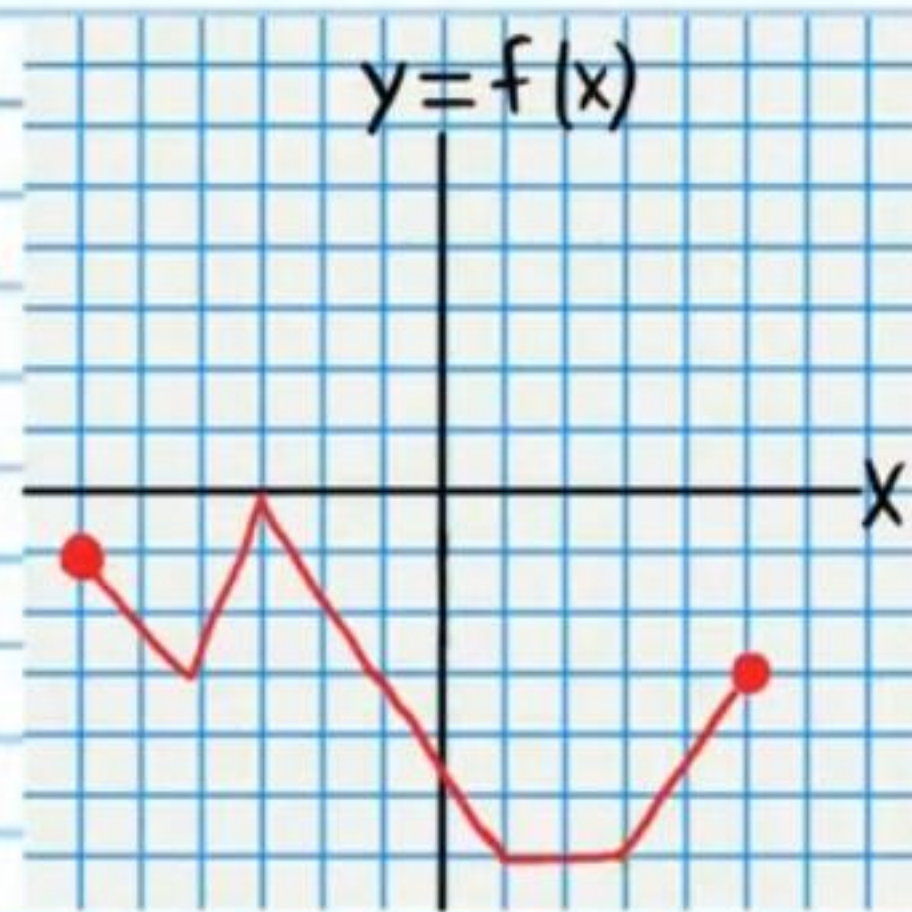


⑩ Domain

Words: All real numbers between -6 and 5, inclusive.

Set Notation: $\{x \mid -6 \leq x \leq 5\}$

Interval Notation: $[-6, 5]$



Range

Words: All real numbers between -6 and zero, inclusive.

Set Notation: $\{y \mid -6 \leq y \leq 0\}$

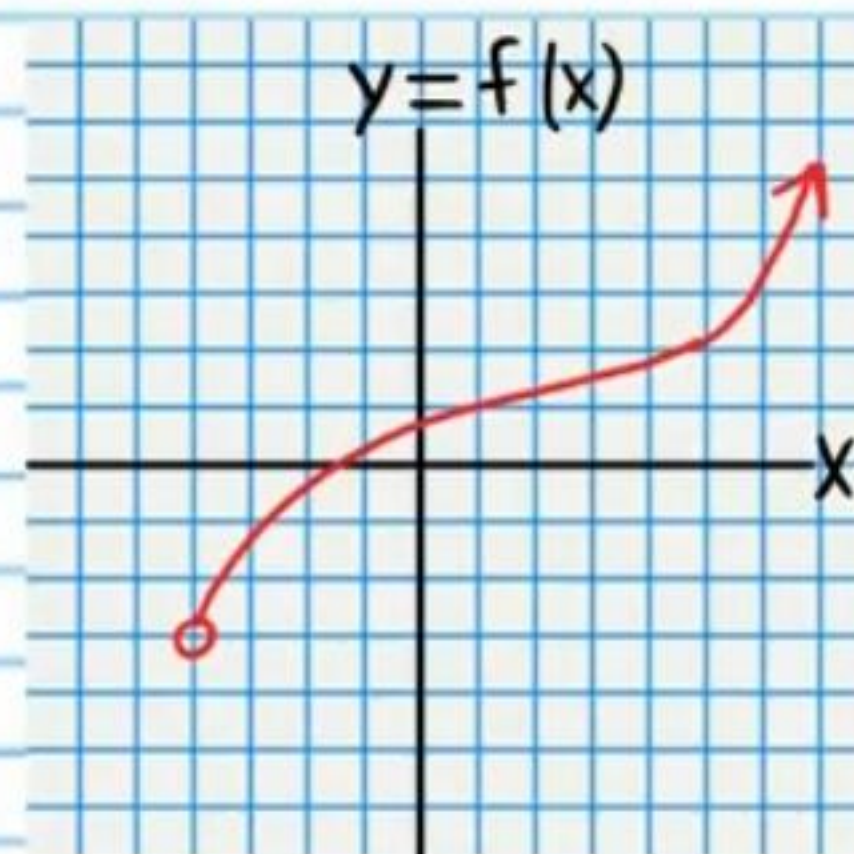
Interval Notation: $[-6, 0]$

⑪ Domain

Words: All real numbers greater than -4.

Set Notation: $\{x \mid x > -4\}$

Interval Notation: $(-4, \infty)$



Range

Words: All real numbers greater than -3.

Set Notation: $\{y \mid y > -3\}$

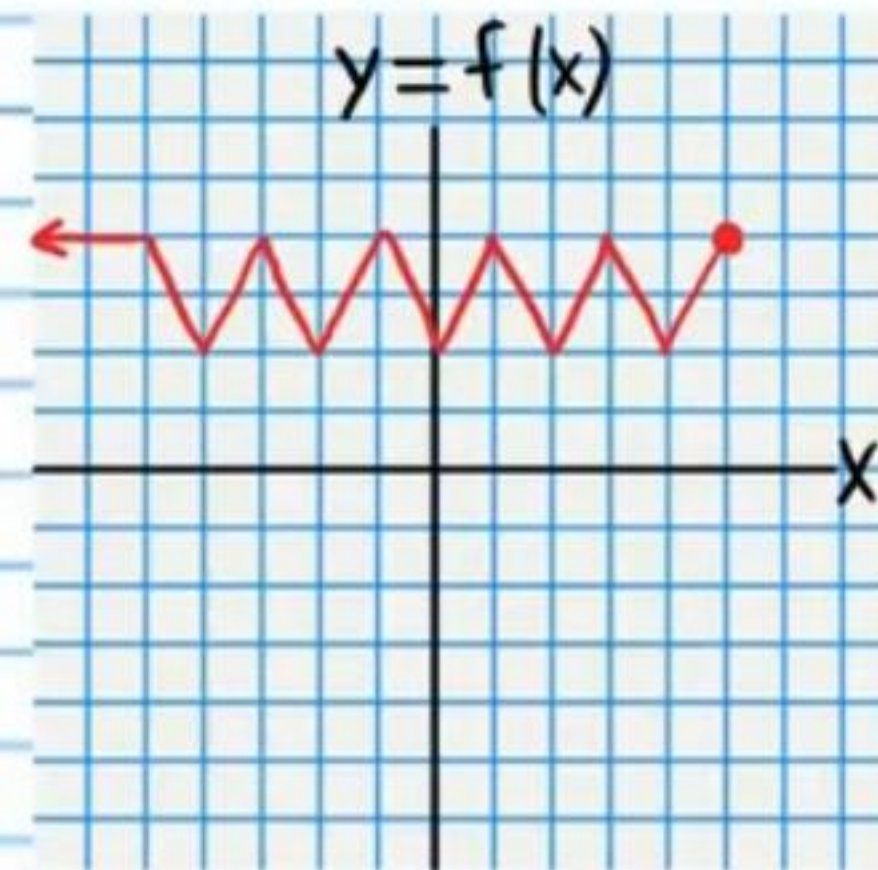
Interval Notation: $(-3, \infty)$

⑫ Domain

Words: All real numbers less than or equal to 5.

Set Notation: $\{x \mid x \leq 5\}$

Interval Notation: $(-\infty, 5]$



Range

Words: All real numbers between 2 and 4, inclusive.

Set Notation: $\{y \mid 2 \leq y \leq 4\}$

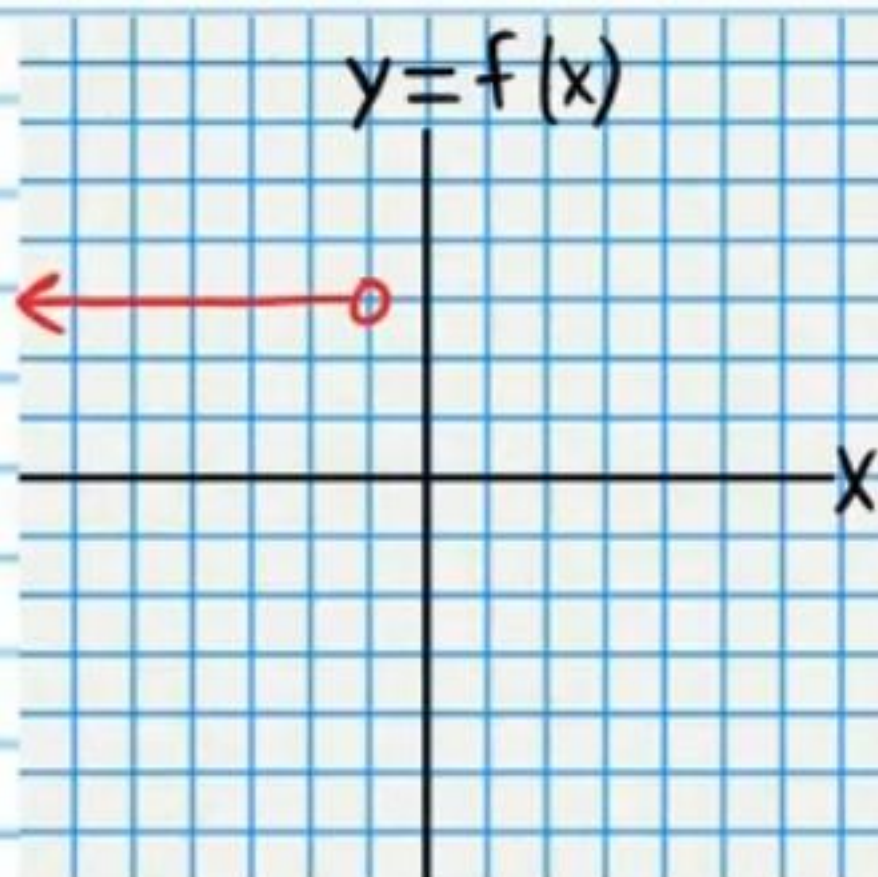
Interval Notation: $[2, 4]$

⑬ Domain

Words: All real numbers less than -1.

Set Notation: $\{x \mid x < -1\}$

Interval Notation: $(-\infty, -1)$



Range

Words: The only value in the range is 3.

Set Notation: $\{y \mid y = 3\}$

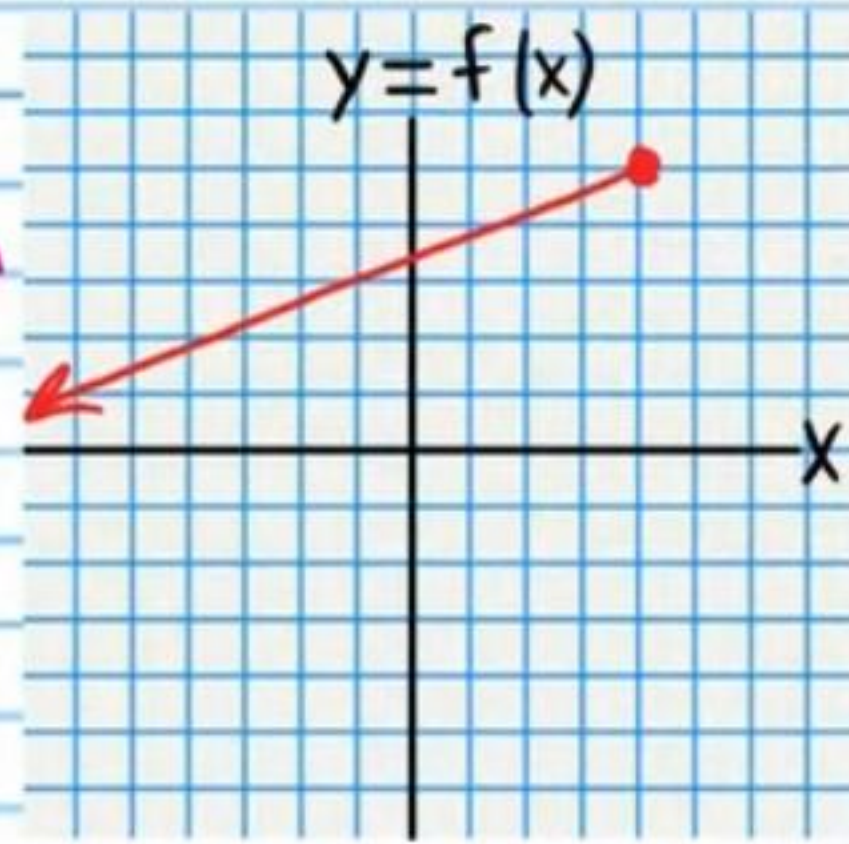
Interval Notation: $[3, 3]$

⑭ Domain

Words: All real numbers less than or equal to 4.

Set Notation: $\{x | x \leq 4\}$

Interval Notation: $(-\infty, 4]$



Range

Words: All real numbers less than or equal to 5.

Set Notation: $\{y | y \leq 5\}$

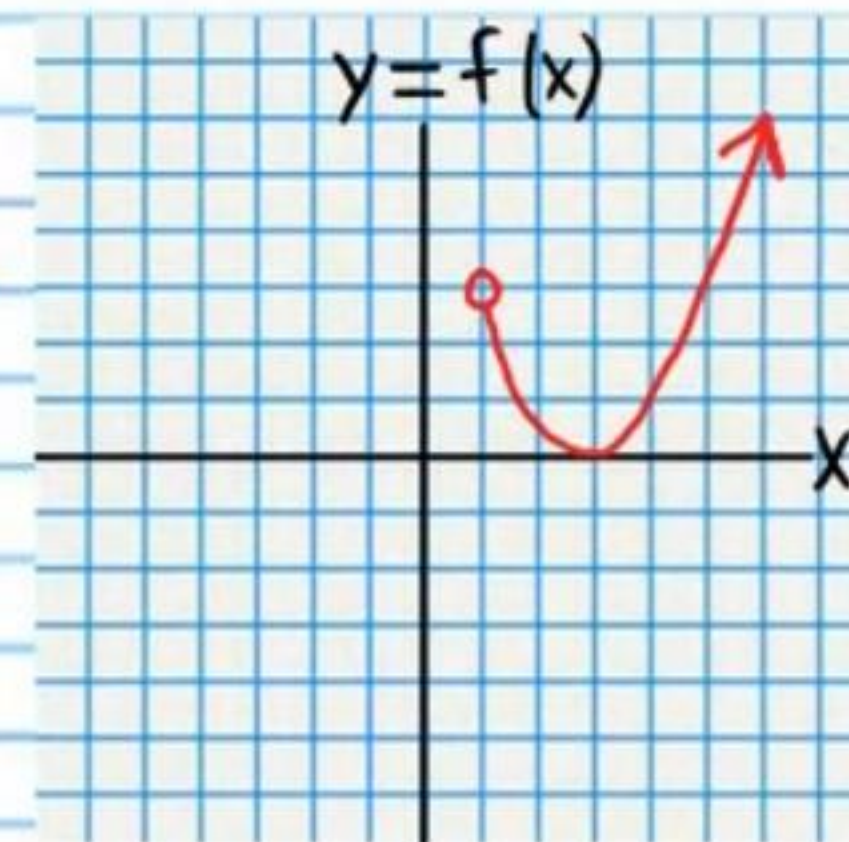
Interval Notation: $(-\infty, 5]$

⑮ Domain

Words: All real numbers greater than 1.

Set Notation: $\{x | x > 1\}$

Interval Notation: $(1, \infty)$



Range

Words: All real numbers greater than or equal to zero.

Set Notation: $\{y | y \geq 0\}$

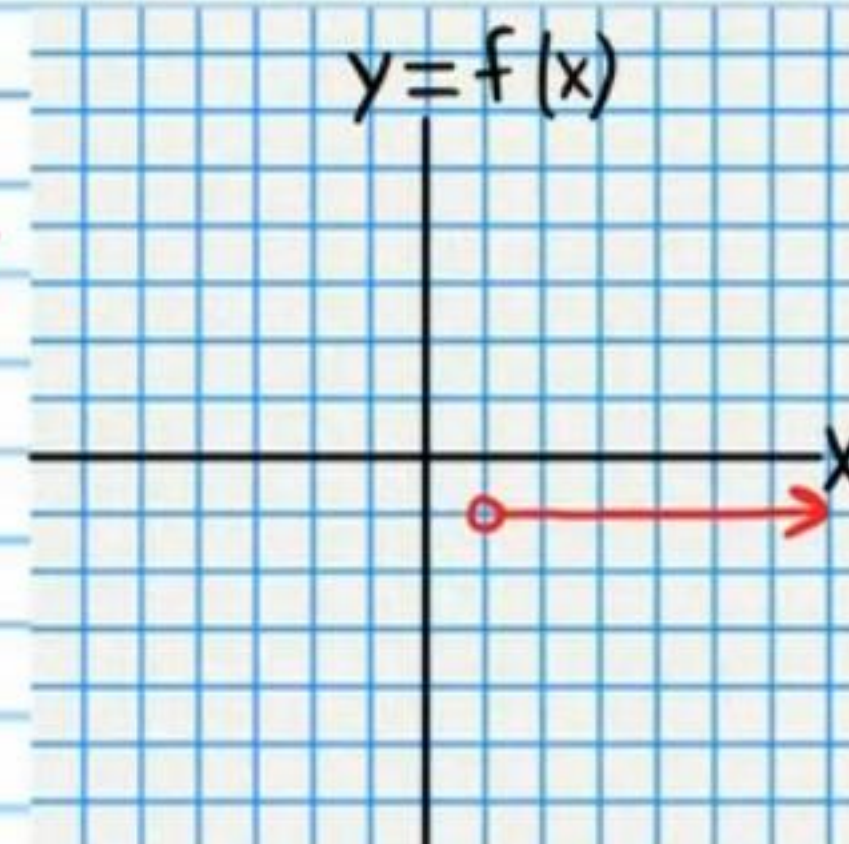
Interval Notation: $[0, \infty)$

⑯ Domain

Words: All real numbers greater than 1.

Set Notation: $\{x | x > 1\}$

Interval Notation: $(1, \infty)$



Range

Words: The only value in the range is -1.

Set Notation: $\{y | y = -1\}$

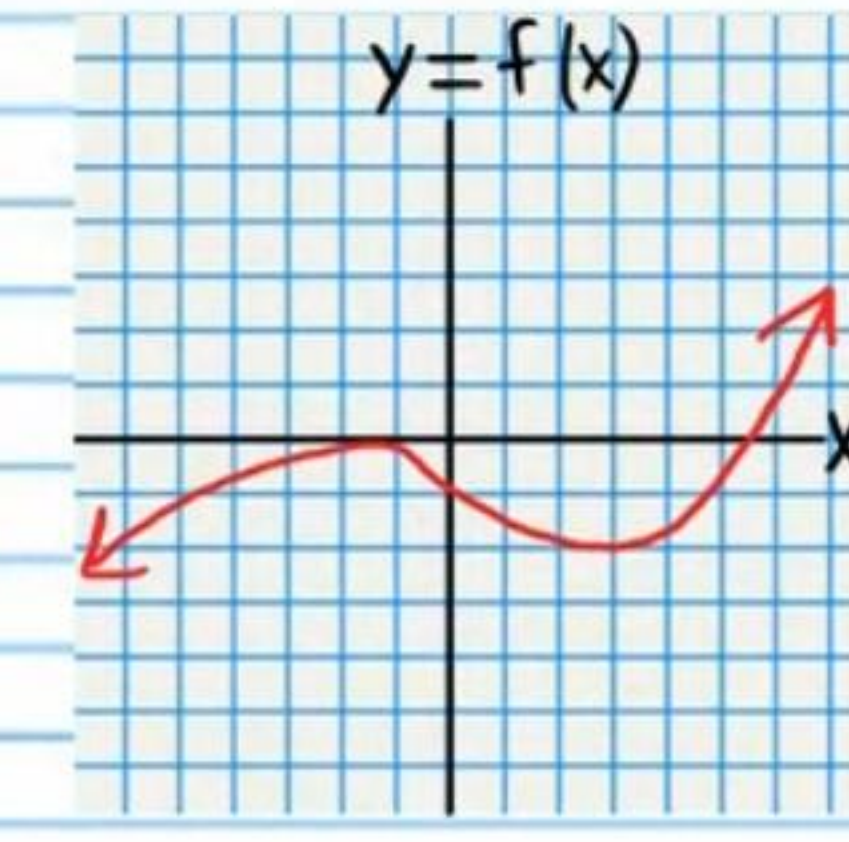
Interval Notation: $[-1, -1]$

⑰ Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$



Range

Words: All real numbers.

Set Notation: All real numbers

Interval Notation: $(-\infty, \infty)$

⑮ Domain

Words: All real numbers.

Set Notation: All real numbers.

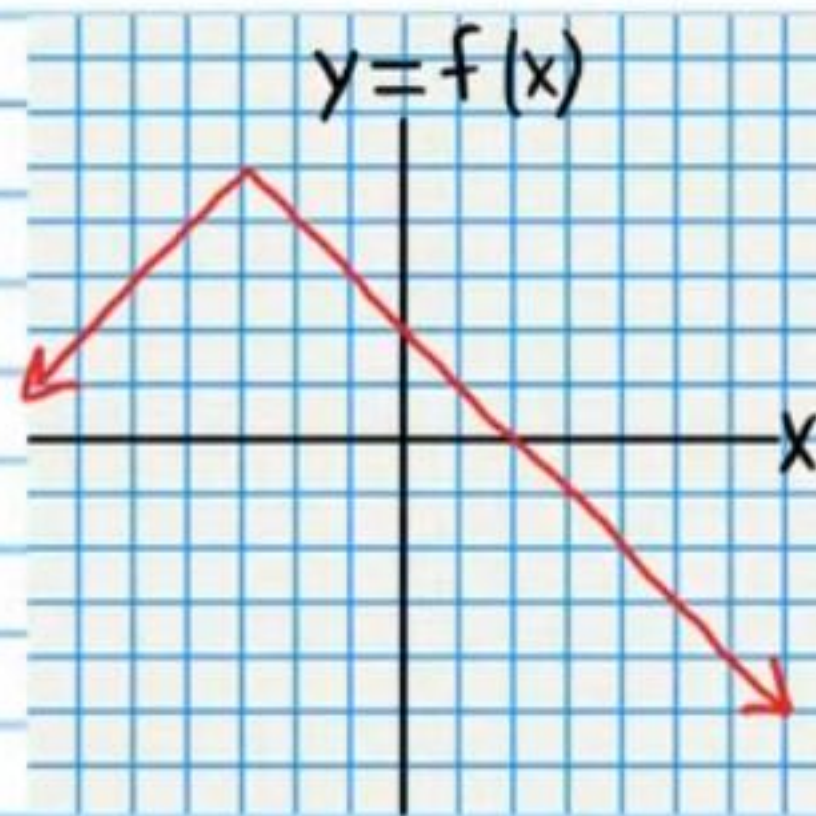
Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers less than or equal to 5.

Set Notation: $\{y \mid y \leq 5\}$

Interval Notation: $(-\infty, 5]$



⑯ Domain

Words: All real numbers.

Set Notation: All real numbers.

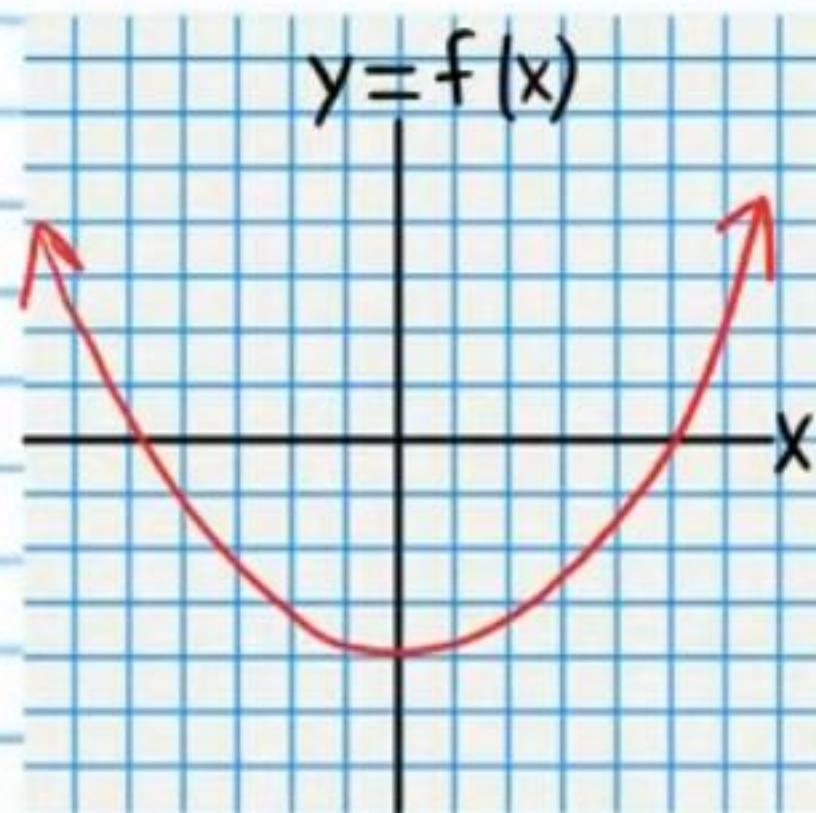
Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers greater than or equal to -4.

Set Notation: $\{y \mid y \geq -4\}$

Interval Notation: $[-4, \infty)$



⑰ Domain

Words: All real numbers. * The pattern continues on both sides.

Set Notation: All real numbers.

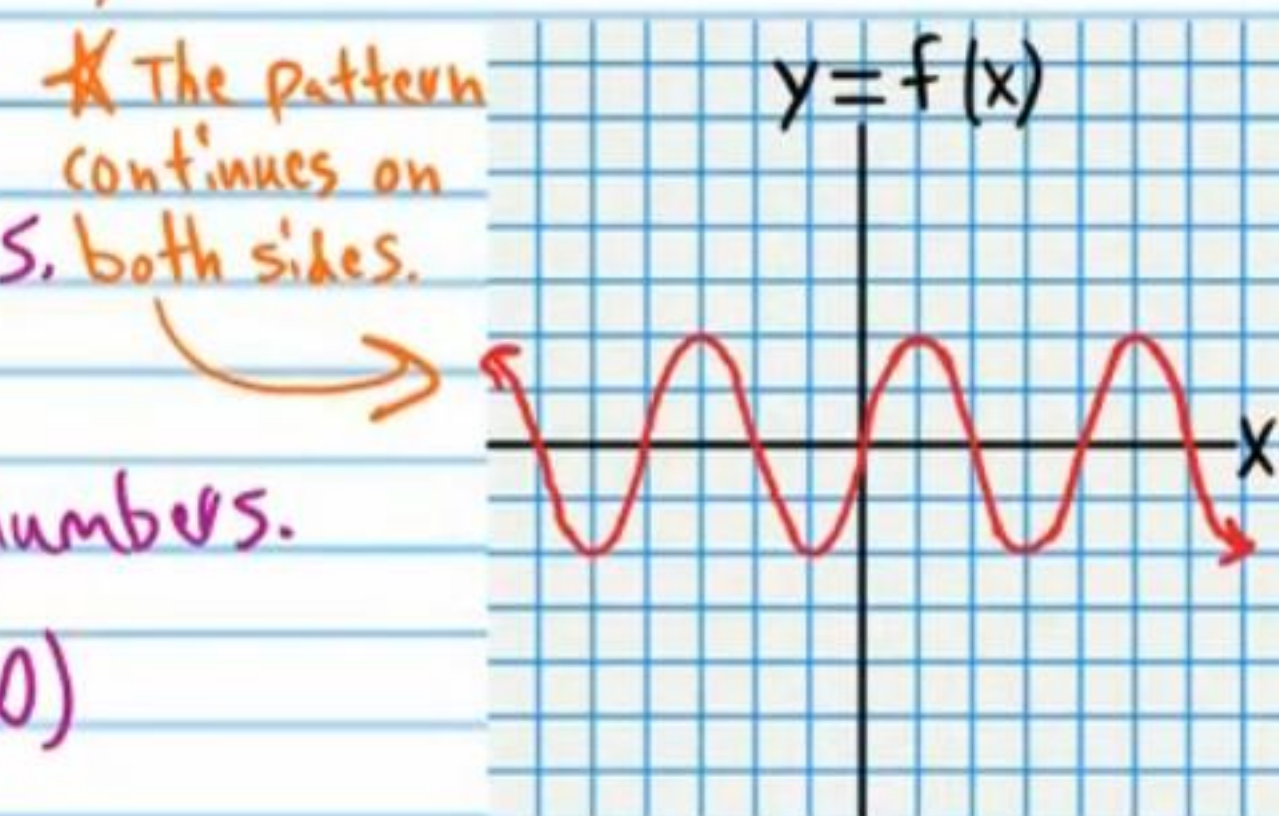
Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers between -2 and 2, inclusive.

Set Notation: $\{y \mid -2 \leq y \leq 2\}$

Interval Notation: $[-2, 2]$



⑱ Domain

Words: All real numbers less than -3 or greater than 1.

Set Notation: $\{x \mid x < -3 \text{ or } x > 1\}$

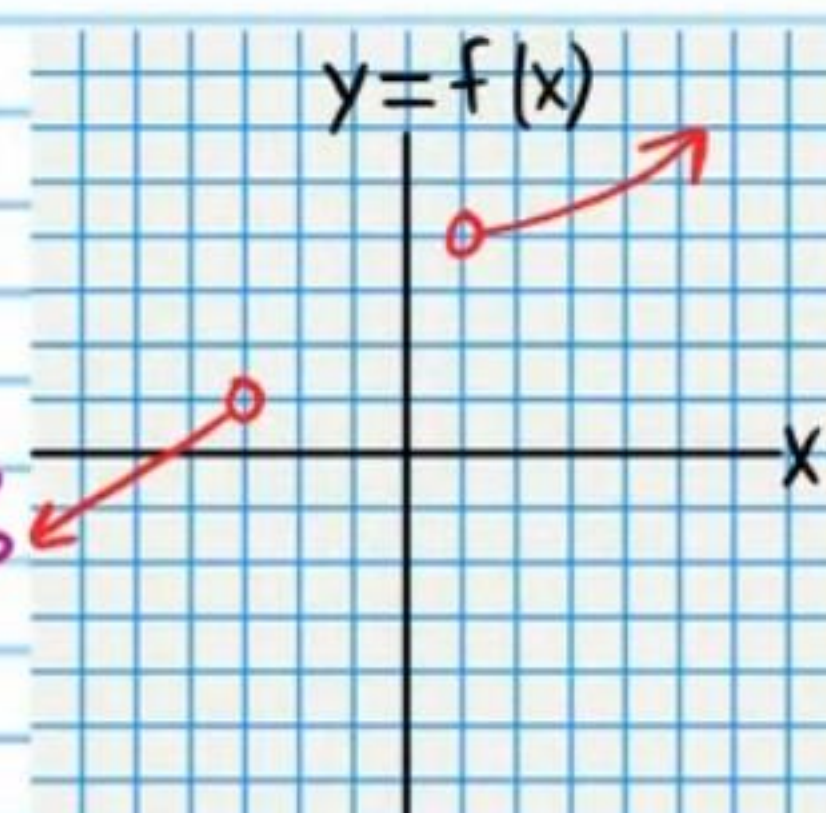
Interval Notation: $(-\infty, -3) \cup (1, \infty)$

Range

Words: All real numbers less than 1 or greater than 4.

Set Notation: $\{y \mid y < 1 \text{ or } y > 4\}$

Interval Notation: $(-\infty, 1) \cup (4, \infty)$



②② Domain

Words: All real numbers less than 2 or greater than 4.

Set Notation: $\{x \mid x < 2 \text{ or } x > 4\}$

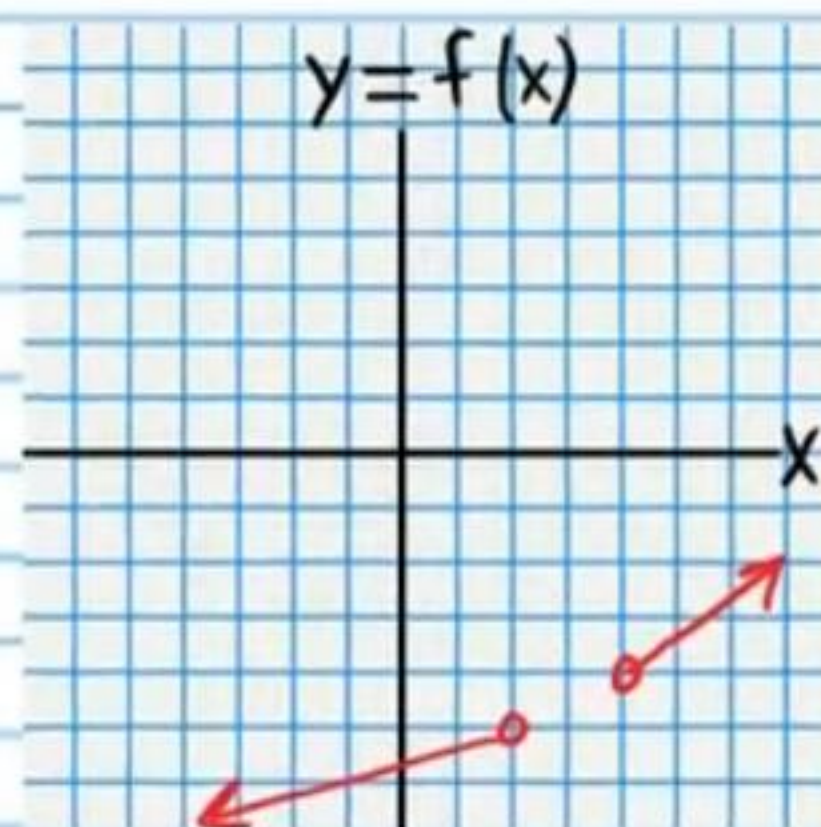
Interval Notation: $(-\infty, 2) \cup (4, \infty)$

Range

Words: All real numbers less than -5 or greater than -4.

Set Notation: $\{y \mid y < -5 \text{ or } y > -4\}$

Interval Notation: $(-\infty, -5) \cup (-4, \infty)$



②③ Domain

Words: All real numbers less than 1 or greater than 3.

Set Notation: $\{x \mid x < 1 \text{ or } x > 3\}$

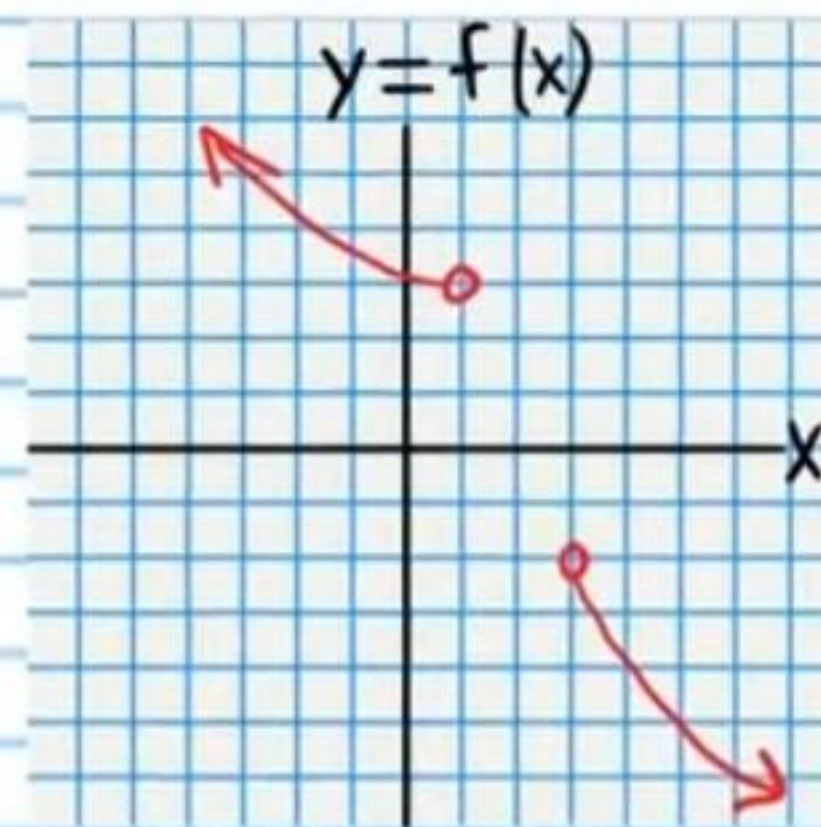
Interval Notation: $(-\infty, 1) \cup (3, \infty)$

Range

Words: All real numbers less than -2 or greater than 3.

Set Notation: $\{y \mid y < -2 \text{ or } y > 3\}$

Interval Notation: $(-\infty, -2) \cup (3, \infty)$



②④ Domain

Words: All real numbers less than 1 or greater than 2.

Set Notation: $\{x \mid x < 1 \text{ or } x > 2\}$

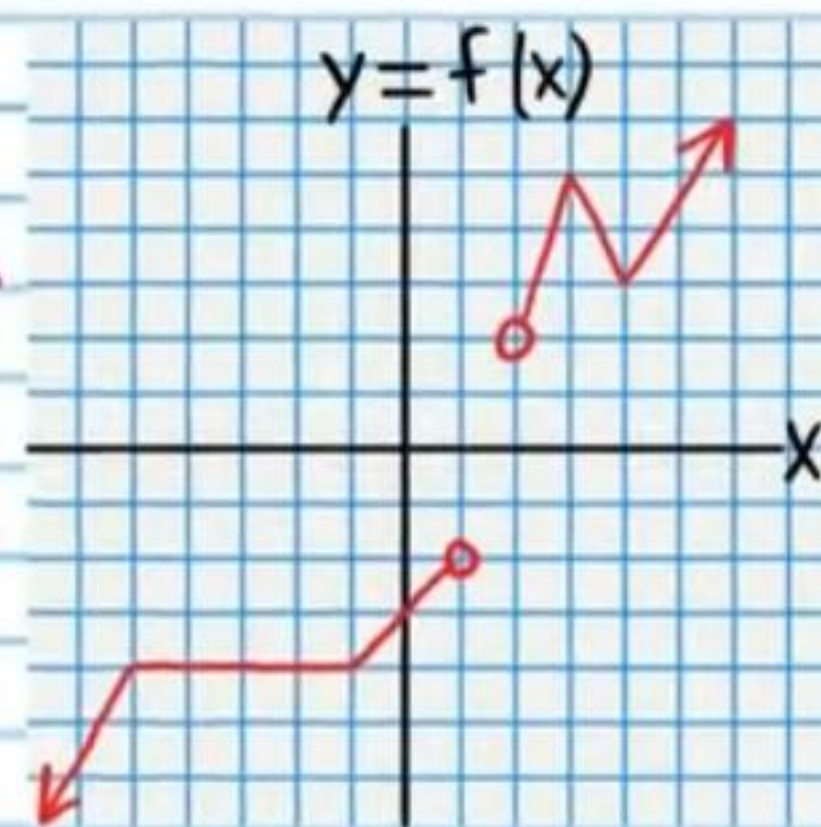
Interval Notation: $(-\infty, 1) \cup (2, \infty)$

Range

Words: All real numbers less than -2 or greater than 2.

Set Notation: $\{y \mid y < -2 \text{ or } y > 2\}$

Interval Notation: $(-\infty, -2) \cup (2, \infty)$



②⑤ Domain

Words: All real numbers except 4.

Set Notation: $\{x \mid x \neq 4\}$

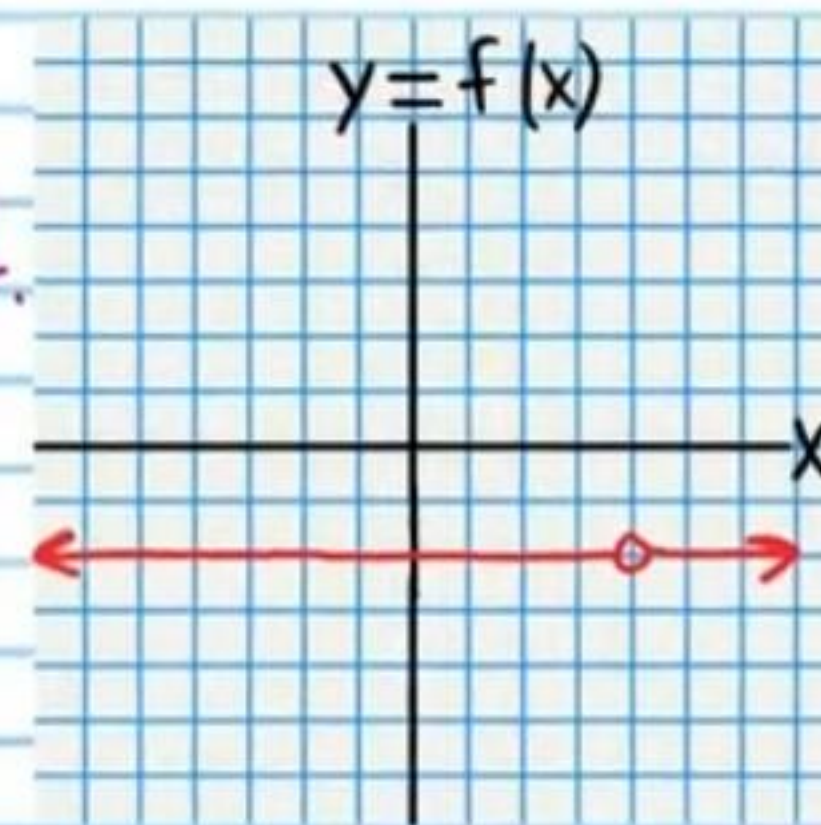
Interval Notation: $(-\infty, 4) \cup (4, \infty)$

Range

Words: The only value in the range is -2.

Set Notation: $\{y \mid y = -2\}$

Interval Notation: $[-2, -2]$

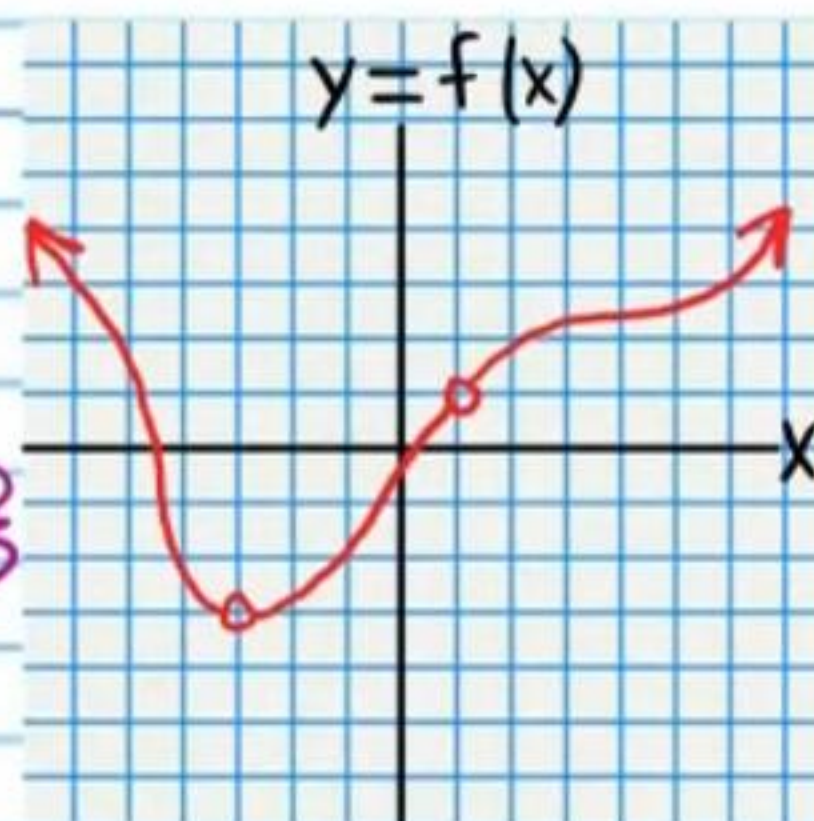


②⑥ Domain

Words: All real numbers except -3 or 1.

Set Notation: $\{x \mid x \neq -3 \text{ and } x \neq 1\}$

Interval Notation: $(-\infty, -3) \cup (-3, 1) \cup (1, \infty)$



Range

Words: All real numbers greater than -3.

Set Notation: $\{y \mid y > -3\}$

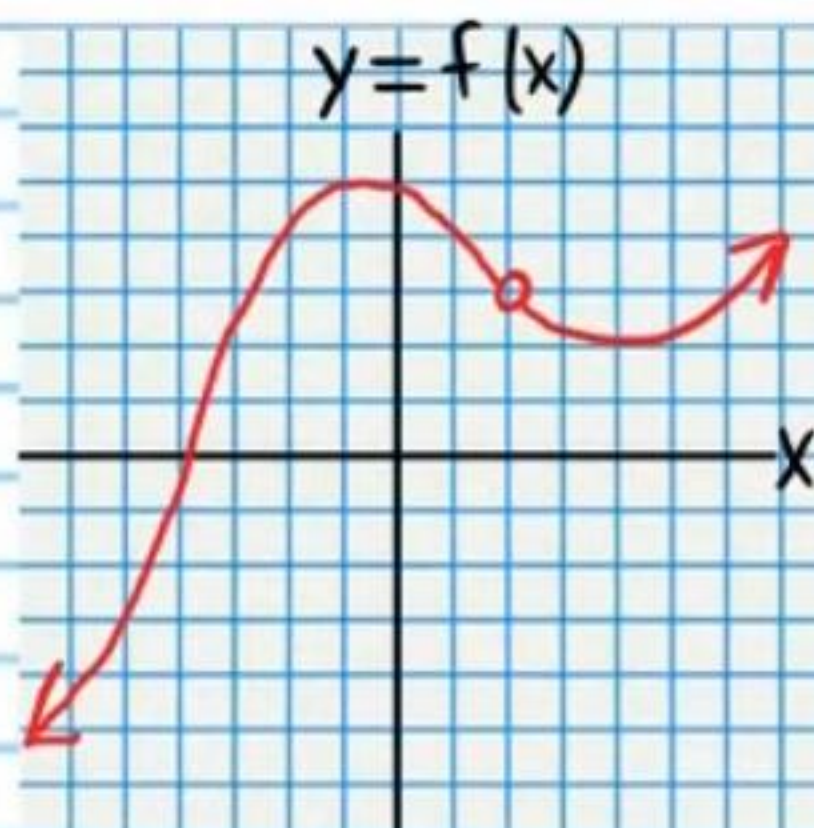
Interval Notation: $(-3, \infty)$

②⑦ Domain

Words: All real numbers except 2.

Set Notation: $\{x \mid x \neq 2\}$

Interval Notation: $(-\infty, 2) \cup (2, \infty)$



Range

Words: All real numbers.

Set Notation: All real numbers.

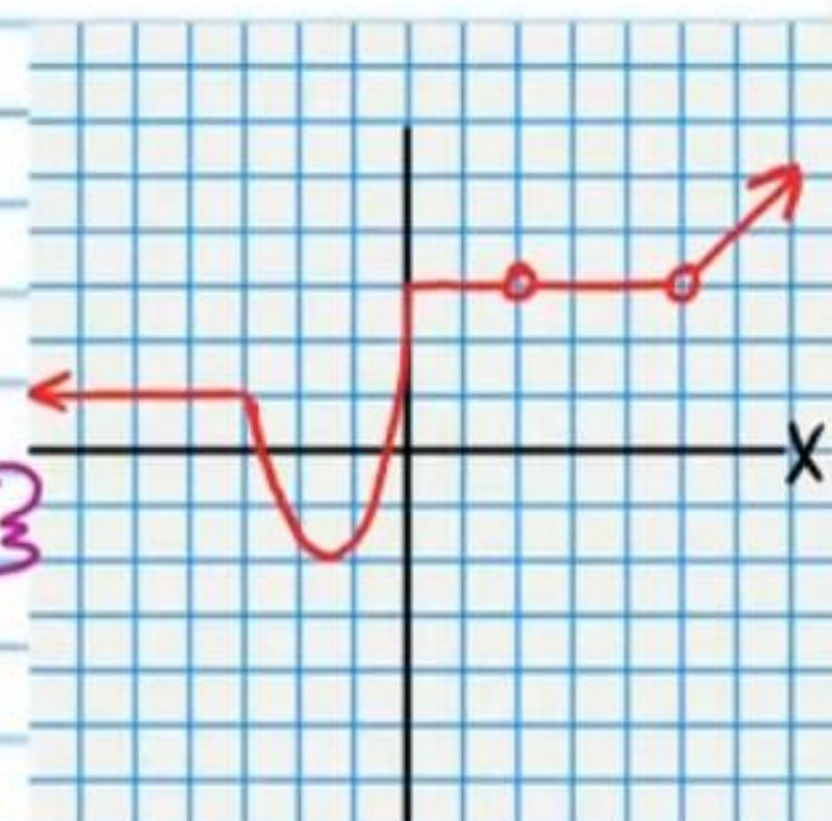
Interval Notation: $(-\infty, \infty)$

②⑧ Domain

Words: All real numbers except 2 and 5.

Set Notation: $\{x \mid x \neq 2 \text{ and } x \neq 5\}$

Interval Notation: $(-\infty, 2) \cup (2, 5) \cup (5, \infty)$



Range

Words: All real numbers greater than or equal to -2.

Set Notation: $\{y \mid y \geq -2\}$

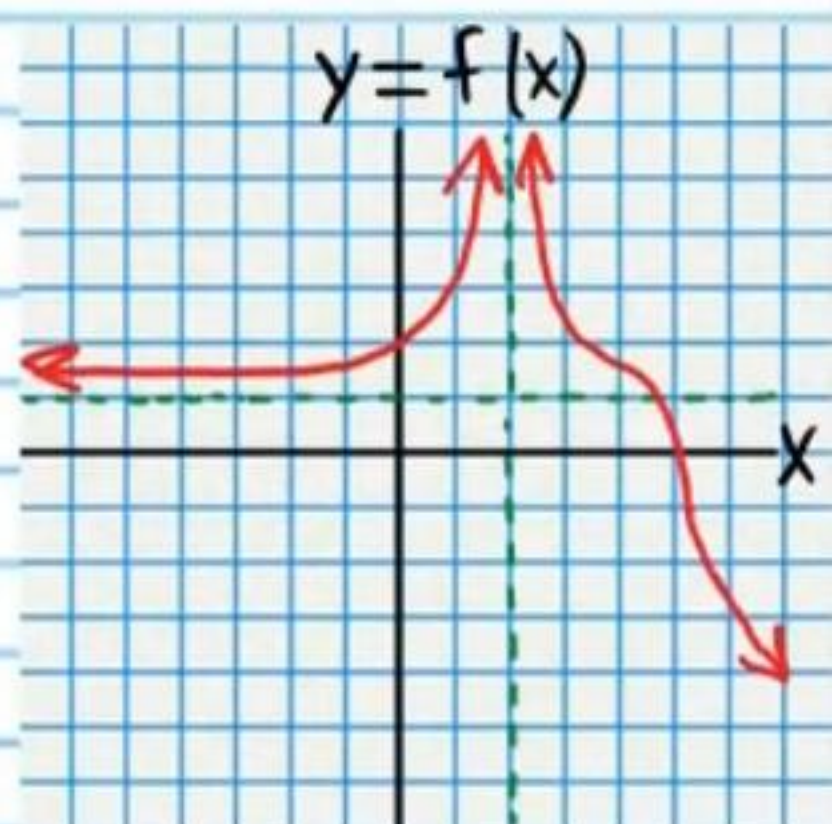
Interval Notation: $[-2, \infty)$

②⑨ Domain

Words: All real numbers except 2.

Set Notation: $\{x \mid x \neq 2\}$

Interval Notation: $(-\infty, 2) \cup (2, \infty)$



Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

③① Domain

Words: All real numbers except -4 and 3.

Set Notation: $\{x \mid x \neq -4 \text{ and } x \neq 3\}$

Interval Notation: $(-\infty, -4) \cup (-4, 3) \cup (3, \infty)$

Range

Words: All real numbers except -2.

Set Notation: $\{y \mid y \neq -2\}$

Interval Notation: $(-\infty, -2) \cup (-2, \infty)$

③② Domain

Words: All real numbers except -3.

Set Notation: $\{x \mid x \neq -3\}$

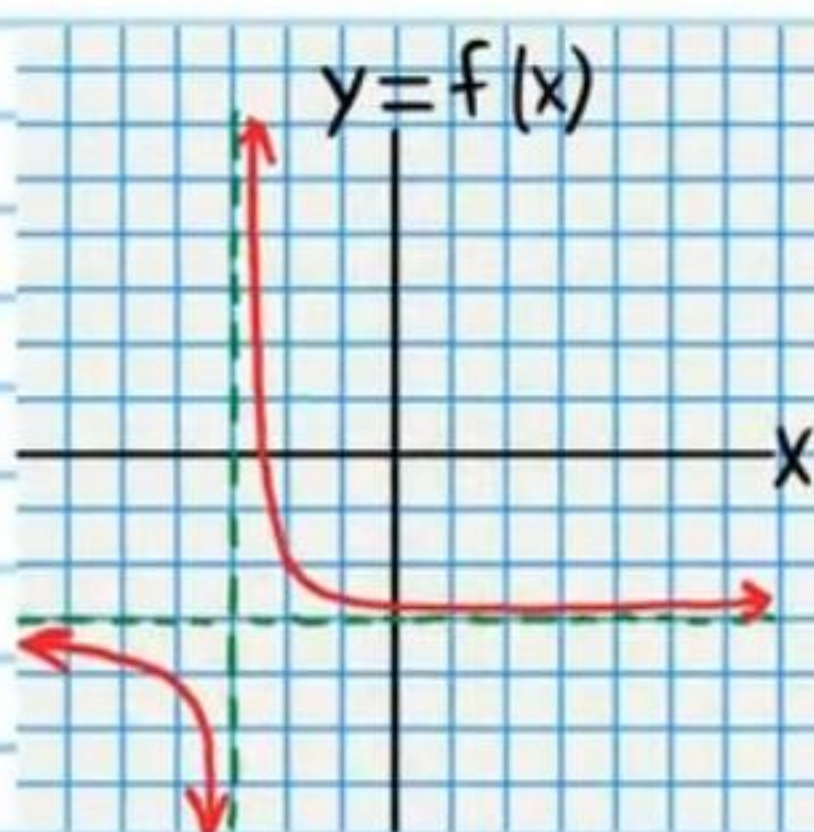
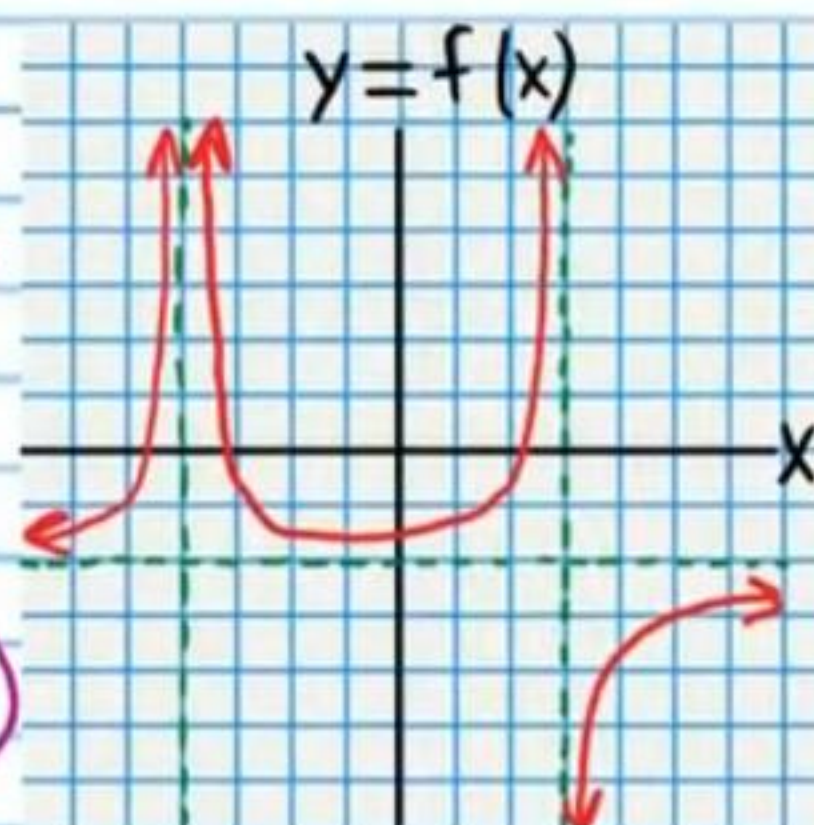
Interval Notation: $(-\infty, -3) \cup (-3, \infty)$

Range

Words: All real numbers except -3.

Set Notation: $\{y \mid y \neq -3\}$

Interval Notation: $(-\infty, -3) \cup (-3, \infty)$



③② Domain

Words: All real numbers except -5 and 2.

Set Notation: $\{x \mid x \neq -5 \text{ and } x \neq 2\}$

Interval Notation: $(-\infty, -5) \cup (-5, 2) \cup (2, \infty)$

Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

③③ Domain

Words: All real numbers.

Set Notation: All real numbers

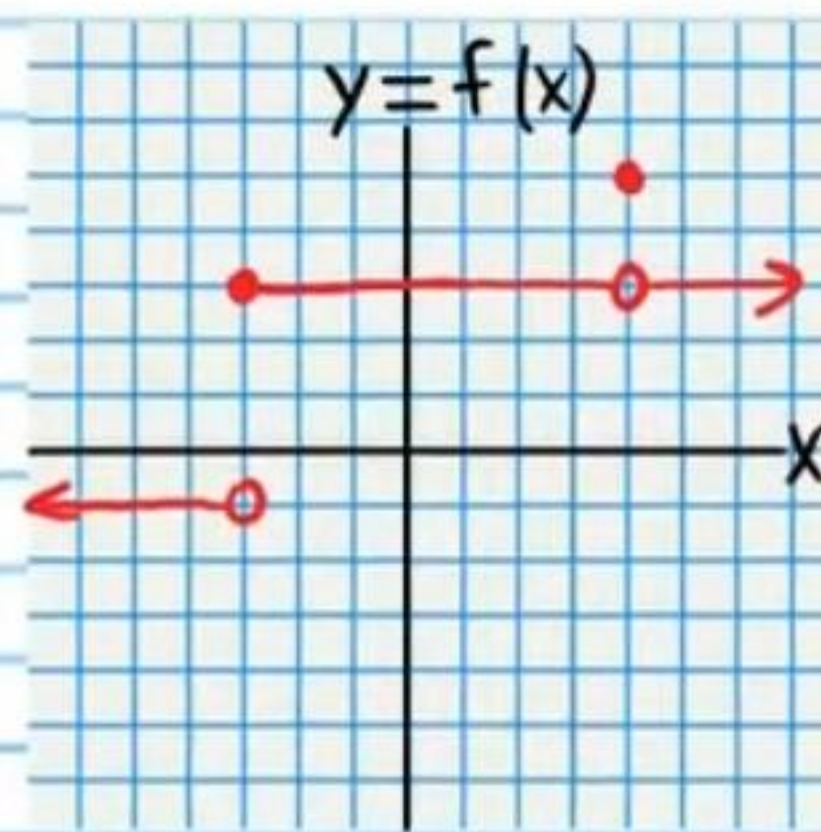
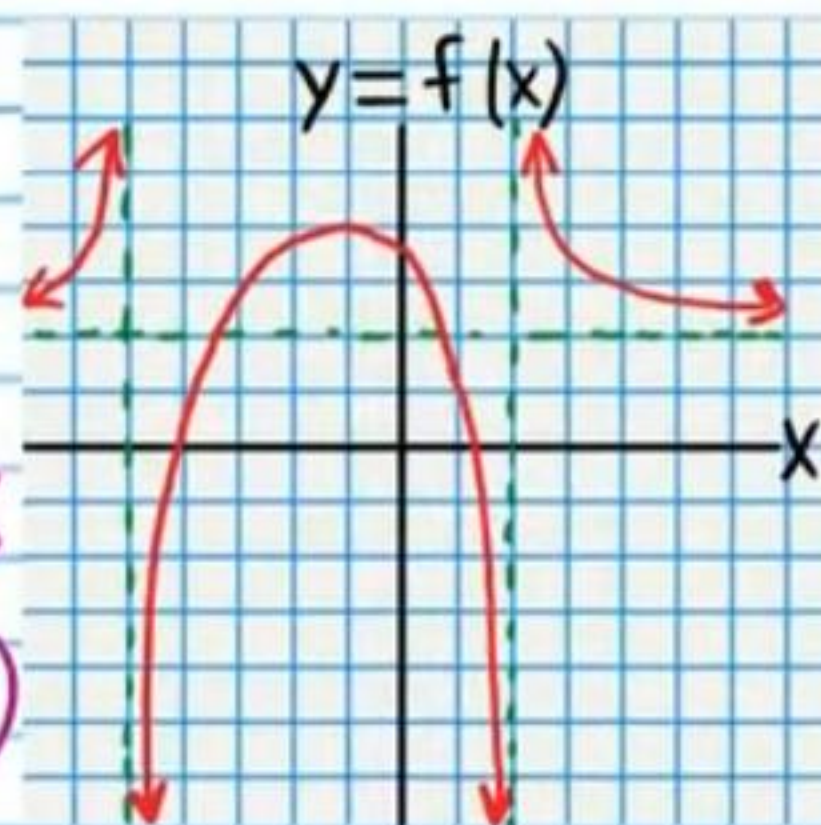
Interval Notation: $(-\infty, \infty)$

Range

Words: The only numbers in the range are -1, 3, and 5.

Set Notation: $\{y \mid y = -1, 3, \text{ or } 5\}$

Interval Notation: $[-1, -1] \cup [3, 3] \cup [5, 5]$



③④ Domain

Words: All real numbers except -4.

Set Notation: $\{x \mid x \neq -4\}$

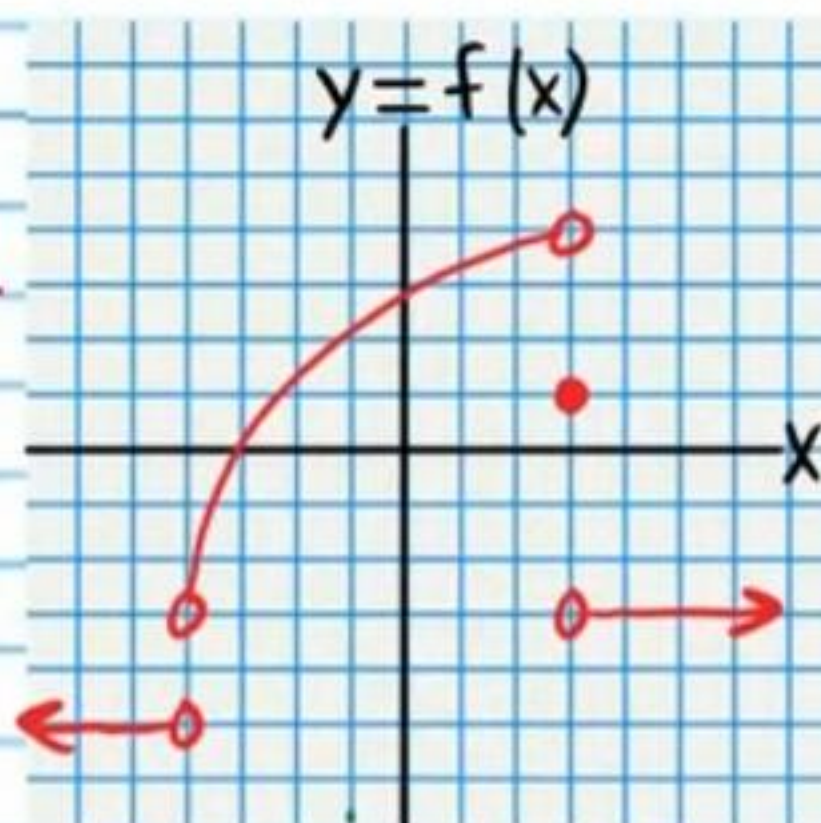
Interval Notation: $(-\infty, -4) \cup (-4, \infty)$

Range

Words: -5 or all real numbers between -3 and 4, including -3.

Set Notation: $\{y \mid y = -5 \text{ or } -3 \leq y < 4\}$

Interval Notation: $[-5, -5] \cup [-3, 4)$



③⑤ Domain

Words: The only numbers in the domain are -3, 2, 3, and 4.

Set Notation: $\{x \mid x = -3, 2, 3, \text{ or } 4\}$

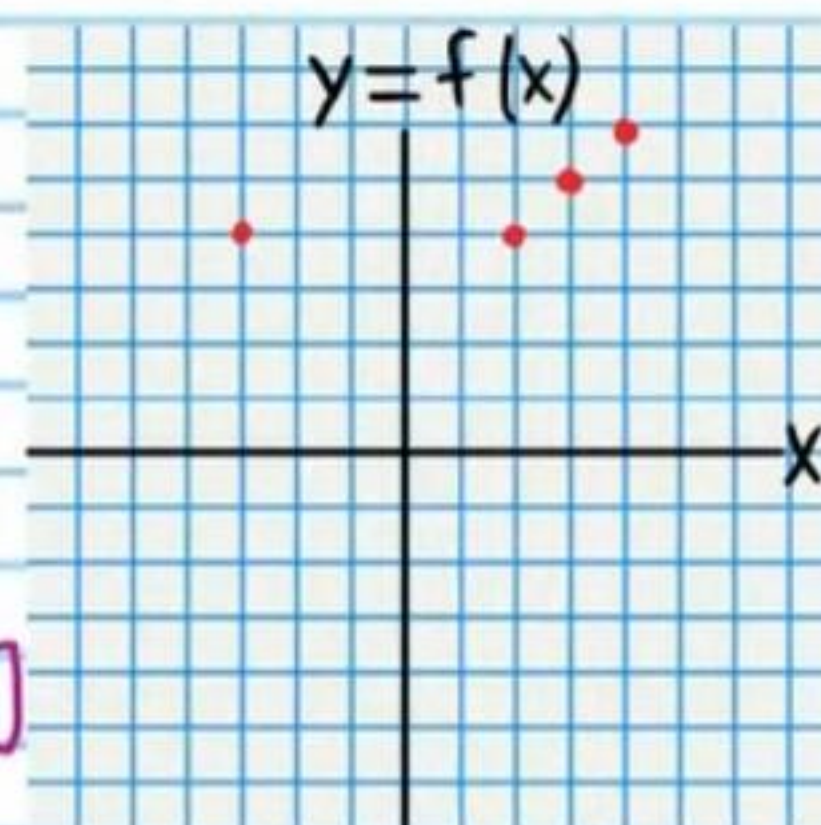
Interval Notation: $[-3, -3] \cup [2, 2] \cup [3, 3] \cup [4, 4]$

Range

Words: The only values in the range are 4, 5, and 6.

Set Notation: $\{y \mid y = 4, 5, \text{ or } 6\}$

Interval Notation: $[4, 4] \cup [5, 5] \cup [6, 6]$



③⑥ Domain

Words: All real numbers except 3.

Set Notation: $\{x \mid x \neq 3\}$

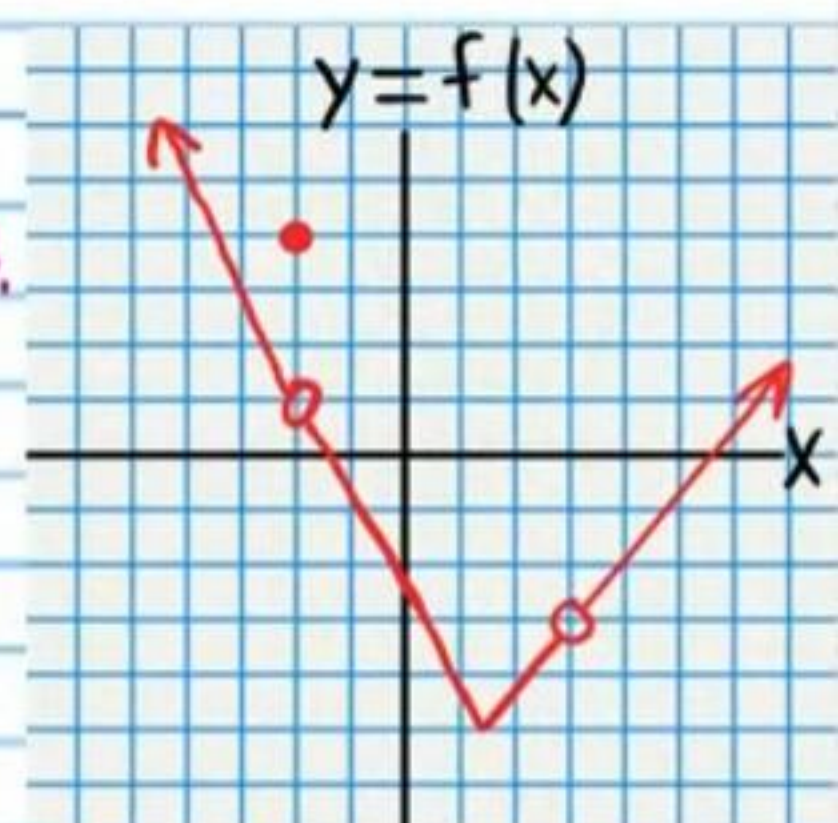
Interval Notation: $(-\infty, 3) \cup (3, \infty)$

Range

Words: All real numbers greater than or equal to -5.

Set Notation: $\{y \mid y \geq -5\}$

Interval Notation: $[-5, \infty)$



③⑦ Domain

Words: All real numbers.

Set Notation: All real numbers

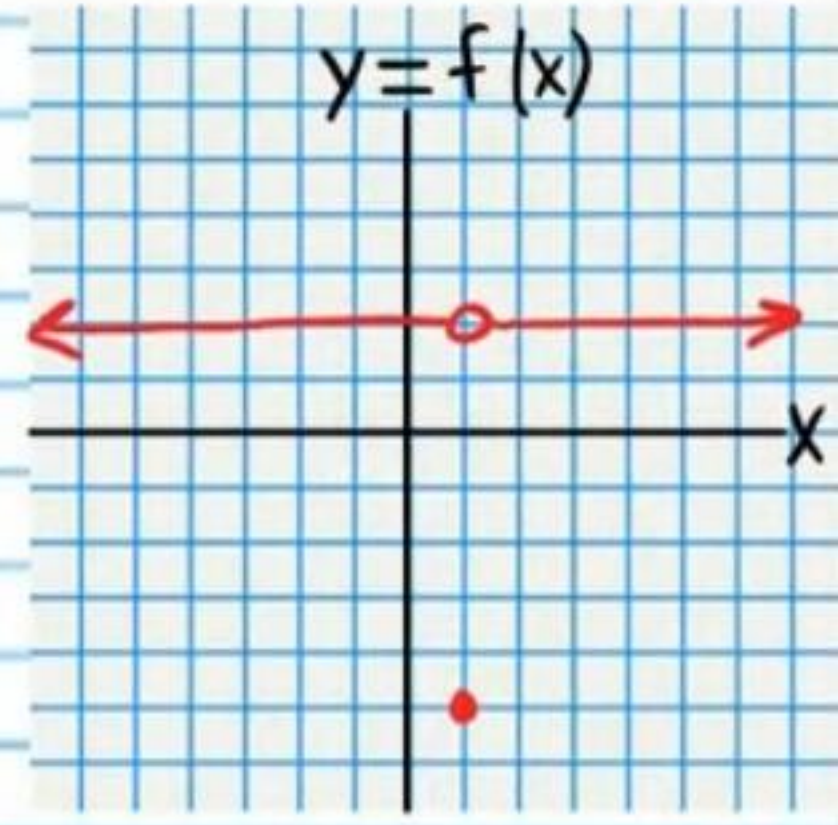
Interval Notation: $(-\infty, \infty)$

Range

Words: The only values in the range are -5 and 2.

Set Notation: $\{y \mid y = -5 \text{ or } y = 2\}$

Interval Notation: $[-5, -5] \cup [2, 2]$

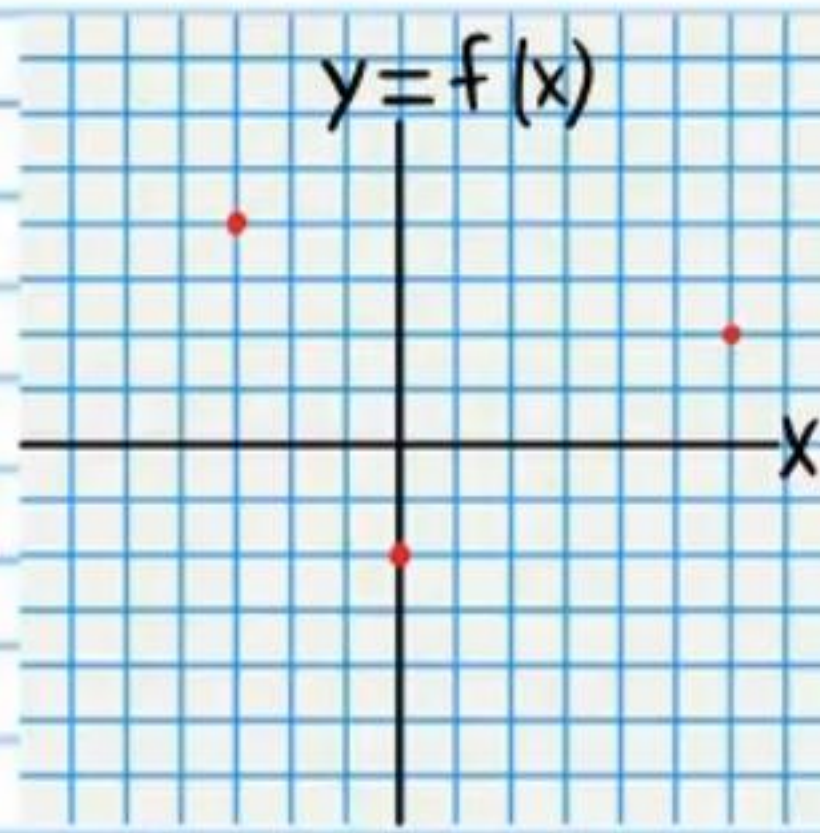


③⑧ Domain

Words: The only values in the domain are -3 , zero, and 6 .

Set Notation: $\{x \mid x = -3, 0, \text{ or } 6\}$

Interval Notation: $[-3, -3] \cup [0, 0] \cup [6, 6]$



Range

Words: The only values in the range are -2 , 2 , and 4 .

Set Notation: $\{y \mid y = -2, 2, \text{ or } 4\}$

Interval Notation: $[-2, -2] \cup [2, 2] \cup [4, 4]$